
Assumptions: EV is charged at home
 Gasoline price = \$3.62/gallon
 Electricity price = \$0.18/kWh
 CO₂ grid intensity = 91.424 kg CO₂/MWh
 = 0.4134 kg CO₂/kWh
 ICE fuel economy = 27.1 mpg
 Hybrid fuel economy = 50 mpg
 EV energy use = 39 kWh/100mi
 = 0.39 kWh/mi
 Tailpipe emissions = 91 kg CO₂/gal

For ICE power train
 gallons used for the 10,000 miles
 1 gal $\rightarrow$ 27.1 mi
 x gal $\rightarrow$ 10,000 mi
 x gal = $\frac{10,000 \text{ mi} \times x \text{ gal}}{27.1}$
 = 369 gal

Use of fuel

\$3.62 $\rightarrow$ 1 gal
 x $\rightarrow$ 369 gal
 = \$1,335.78

Emissions

9 kg CO₂ per gal

$$\begin{array}{ccc} 9 \text{ kg} & \times & 1 \text{ gal} \\ x \text{ kg} & \Rightarrow & 269 \text{ gal} \end{array}$$

$$= 3,321 \text{ kg CO}_2$$

Hybrid power train

$$\begin{array}{ccc} 1 \text{ gal} & \rightarrow & 50 \text{ mi} \\ x \text{ gal} & \rightarrow & 10,000 \text{ mi} \\ & & 200 \end{array}$$

$$x \text{ gal} = \frac{10,000 \text{ mi}}{50 \text{ mi}} = 200 \text{ gal}$$

Fuel cost

$$\begin{array}{ccc} \$ 8.62 & \times & 1 \text{ gal} \\ \$ x & \Rightarrow & 200 \text{ gal} \end{array}$$

$$\$ x = \$ 1724$$

CO₂

$$\begin{array}{ccc} 9 \text{ kg CO}_2 \text{ eq} & \Rightarrow & 1 \text{ gal} \\ x & \rightarrow & 200 \text{ gal} \end{array}$$

$$= \underline{\underline{1,800 \text{ kg CO}_2}}$$

Electric Vehicle Power from
Electricity Consumption

$$\begin{aligned} 0.39 \text{ kWh} &\rightarrow 1 \text{ mi} \\ \times &\quad \rightarrow 10,000 \\ &= 3,900 \text{ kWh} \end{aligned}$$

Cost of energy (Charging at home)

$$\begin{aligned} \$0.18 &\rightarrow 1 \text{ kWh} \\ \times &\rightarrow 3,900 \text{ kWh} \\ &= \$702 \end{aligned}$$

Emissions

$$\begin{aligned} 0.4134 \text{ kg CO}_2 &\rightarrow 1 \text{ kWh} \\ \times &\quad \rightarrow 3,900 \text{ kWh} \\ &= 1,612.3 \text{ kg CO}_2 \end{aligned}$$

Savings

Cost:	ICE	\$ 1,335.78
	hybrid	\$ 724.00
		\$ 611.78

$$= 611.78 \leq \$612$$

$$\begin{array}{r} \text{ICE} \quad \$1325.78 \\ \text{EV} \quad - \quad 702.00 \\ \hline \$633.78 \\ \hline \approx \$634 \end{array}$$

CO₂ Emission savings

$$\begin{array}{r} \text{ICE} \quad 3321 \\ \text{hybrid} \quad 1800 \\ \hline 1,521 \text{ kg CO}_2 \end{array}$$

$$\begin{array}{r} \text{ICE} \quad 3321.0 \\ \text{EV} \quad - \quad 1612.3 \\ \hline 1,708.7 \text{ kg CO}_2 \end{array}$$

This information shows that the EV option offers more economical and environmental benefit. Since it doesn't have tail pipe emissions however, it is important to note that if not being home charged, cost of energy might change from vendor to vendor. Also, the environmental impact of batteries used in both EV & hybrid was not considered

Since the lifecycle phase considered here is only the use phase.

If charging is not done at home, the hybrid option offers a good competitive advantage over the EV. While the savings may vary slightly for EV and hybrid over the given mileage, longer use may show significant savings over time all things being equal.

